Exercise Sheet #4: Electric Field

Discussion: Nov 12, 2025

Exercise 1: Discrete Charge Distribution

Two point charges of $+3.0\,\mu$C each are located at the points $(x, y) = (0, 2.0\,\text{m})$ and $(x, y) = (0, -2.0\,\text{m})$. Two additional point charges, each with charge q, are located at $(x, y) = (4.0\,\text{m}, 2.0\,\text{m})$ and $(x, y) = (4.0\,\text{m}, -2.0\,\text{m})$. The electric field at the point $(x, y) = (0, 0)$ due to the four charges is $\vec{E} = (4.0 \cdot 10^3\ \hat{x})$ N/C. Determine the value of q.

Solution

The electric field at the origin due to a point charge Q at position (x, y) is given by

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\,\hat{r},$$

where $\hat{r}$ is the unit vector pointing from the charge to the field point.
For the two charges at $(0, 2$ m$)$ and $(0, -2$ m$)$, the electric field components at the origin point along the y-axis. Since they are equal in magnitude and opposite in direction, they cancel each other:

$$E_y = 0 \quad \text{(net field from the two } 3.0\,\mu\text{C charges)}$$

The distance to the origin of each q charge is

$$r = \sqrt{(4.0)^2 + (2.0)^2}\,\text{m} = \sqrt{20}\,\text{m},$$

and the electric field at the origin due to one q charge has magnitude

$$E_q = \frac{1}{4\pi\varepsilon_0}\frac{|q|}{r^2}.$$

At the origin, the y-components of the fields from the charges at $(4\,\text{m}, 2\,\text{m})$ and $(4\,\text{m}, -2\,\text{m})$ cancel, while the x-components add.
Thus, the total field at the origin due to both q charges is

$$|E_x| = 2 \cdot E_q \cos\theta,$$

where $\cos\theta = \frac{x}{r} = \frac{4}{4.472} = 0.894$.
Given that the total field at the origin is

$$E_x = 4.0 \cdot 10^3\,\text{N/C},$$

we have

$$q = -\frac{E_x \cdot 4\pi\varepsilon_0 \cdot r^2}{2 \cdot \cos\theta} = -\frac{4.0 \cdot 10^3 \cdot 2\pi \cdot 8.854 \cdot 10^{-12} \cdot 20}{0.894}\,\text{C} = -5.0\,\mu\text{C}.$$

Exercise 2: Electric Displacement

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Solution

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Exercise 3: Gauss's Law

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Solution

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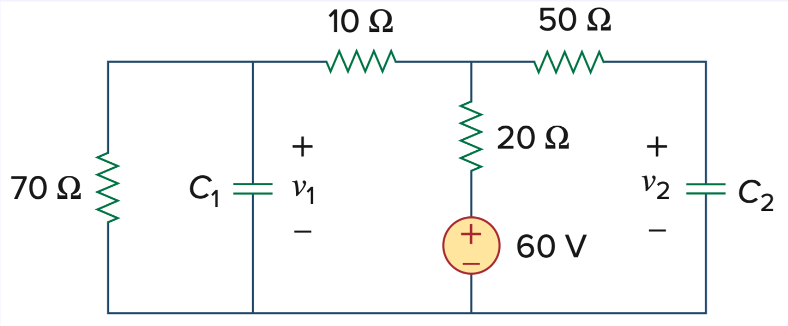
Exercise 4: DC Steady State

Find the voltages across the capacitors in the circuit below under DC conditions.



Solution

Under DC steady-state conditions, the capacitors act as open circuits. Therefore, no current flows through C_1 or C_2, and we can analyze only the resistive network.
Let the bottom node be the reference (0 V). The voltage at the top of the 60 V source is $V_C = 60$ V. Let V_A and V_B be the node voltages at the top of C_1 and the junction between the 10 Ω, 20 Ω, and 50 Ω resistors, respectively.

$$\text{KCL at node } A: \quad \frac{V_A - 0}{70} + \frac{V_A - V_B}{10} + I_{C_1} = 0$$
$$\Rightarrow \frac{V_A - 0}{70} + \frac{V_A - V_B}{10} = 0$$

$$\text{KCL at node } B: \quad \frac{V_B - V_A}{10} + \frac{V_B - 60}{20} + I_{C_2} = 0$$
$$\Rightarrow \frac{V_B - V_A}{10} + \frac{V_B - 60}{20} = 0$$

Multiply the first equation by 70:

$$V_A + 7(V_A - V_B) = 0 \quad \Longrightarrow \quad 8V_A - 7V_B = 0$$

Multiply the second equation by 20:

$$2(V_B - V_A) + (V_B - 60) = 0 \quad \Longrightarrow \quad -2V_A + 3V_B = 60$$

Substitute into each other:

$$-2\left(\frac{7}{8}V_B\right) + 3V_B = 60$$

$$\Rightarrow \left(-\frac{14}{8} + \frac{24}{8}\right) V_B = 60$$

$$\Rightarrow V_B = 60 \cdot \frac{8}{10} = 48 \text{ V}$$

$$V_A = \frac{7}{8} V_B = \frac{7}{8} \cdot 48 = 42 \text{ V} = v_1$$

Finally, let V_D be the trivial node (just one input and one output) voltage between the top of C_2 and the $50\,\Omega$ resistor.

$$\text{KCL at node } D: \quad \frac{V_D - V_B}{50} + I_{C_2} = 0$$

$$I_{C_2} = 0 \text{ (DC steady state)} \;\Rightarrow\; \frac{V_D - V_B}{50} = 0 \;\Rightarrow\; V_D = V_B \;\Rightarrow\; v_2 = V_B$$

$$\boxed{v_1 = 42 \text{ V}, \quad v_2 = 48 \text{ V}}$$

Exercise 5: Plate Capacitor

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Solution

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Exercise 6: ...

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Solution

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Exercise 7: ...

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Solution

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Exercise 8: ...

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Solution

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